Knapsack comparison reference — inputs and method

Typed computational reference — not a handwritten trace. Regenerated in v2.3.1 from the dashboard's v2.3 calculations by `tools/make_knapsack_reference_pdf.py`; it supersedes the typed sheet Knapsack_HandTrace_Sheet_Joan.pdf from the Investment Analyst's (Joan's) Tab 3 pack, which is archived unchanged in `archive/superseded_references/`. The member's genuine handwritten knapsack trace is a separate requirement and is still pending.

Baseline inputs

Threshold X	Hub	Proposed internal hub-to-branch transfer target	First-year investment budget	Eligible routes
38 min (route exists if Minutes < X)	HUB — selected model hub, Al Ramool / Umm Ramool	20 min (direct transfer passes if shortest time ≤ target)	AED 450,000	13

Question

At the 20-minute target, Dijkstra from HUB gives KAR 13, QUS 16, UMS 23, JAF 47, SHJ 24, AJM 51 min: KAR, QUS meet it by direct hub transfer; the other four each have one priced upgrade. With a first-year budget of AED 450,000, which upgrades give the largest modelled annual gross-benefit proxy without exceeding the budget?

Inputs (rows in the group's matrix order)

Row	Office	Upgrade	First-year cost (AED)	Cost (AED 10,000 units)	Annual gross-benefit proxy (AED/yr)	Benefit (AED '000)	Benefit per AED	Service mechanism (assumption)
1	UMS	Express motorbike courier	69,900	7	135,000	135	1.93	Alternative mode: 18 min (assumption, from Al Ramool hub)
2	JAF	JAFZA mini-hub	143,520	14	360,000	360	2.51	Local fulfilment (stock held at branch) — hub travel time unchanged
3	SHJ	Sharjah mini-hub	130,920	13	225,000	225	1.72	Local fulfilment (stock held at branch) — hub travel time unchanged
4	AJM	Ajman mini-hub	137,520	14	180,000	180	1.31	Local fulfilment (stock held at branch) — hub travel time unchanged
	Budget		450,000	45				

Rule for every cell

K[0][w] = 0 for every w. For row i and budget w: if $cost_i$ > w then K[i][w] = K[i–1][w] (cannot afford it); otherwise K[i][w] = max(K[i–1][w] (leave it), $benefit_i$ + K[i–1][w – $cost_i$] (take it)). Each upgrade is taken 0 or 1 times.

Traceback and greedy comparison

Start at K[4][45]. If a cell differs from the one directly above, that upgrade was taken: move left by its cost; otherwise go up a row. Greedy sorts by benefit per AED (highest first) and takes each upgrade that still fits.

Rounding note: costs are rounded to the nearest AED 10,000 only to keep the grid to 46 columns. The dashboard computes in exact AED (unit AED 20); both give the same choice at this budget.

Completed knapsack grid

Cells = best annual gross-benefit proxy in AED '000 using the first i upgrades with budget w (units of AED 10,000). Highlighted cells = traceback path. Bottom-right K[4][45] = **765**.

row / budget w	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
0 (none)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1 UMS (c=7, b=135)	0	0	0	0	**0**	0	0	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135
2 JAF (c=14, b=360)	0	0	0	0	0	0	0	135	135	135	135	135	135	135	360	360	360	360	**360**	360	360	495	495
3 SHJ (c=13, b=225)	0	0	0	0	0	0	0	135	135	135	135	135	135	225	360	360	360	360	360	360	360	495	495
4 AJM (c=14, b=180)	0	0	0	0	0	0	0	135	135	135	135	135	135	225	360	360	360	360	360	360	360	495	495

row / budget w	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
0 (none)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1 UMS (c=7, b=135)	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135
2 JAF (c=14, b=360)	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495
3 SHJ (c=13, b=225)	495	495	495	495	585	585	585	585	**585**	585	585	720	720	720	720	720	720	720	720	720	720	720	720
4 AJM (c=14, b=180)	495	495	495	495	585	585	585	585	585	585	585	720	720	720	720	720	720	720	765	765	765	765	**765**

Traceback: K[4][45] = 765 vs K[3][45] = 720: TAKE AJM · K[3][31] = 585 vs K[2][31] = 495: TAKE SHJ · K[2][18] = 360 vs K[1][18] = 135: TAKE JAF · K[1][4] = 0 vs K[0][4] = 0: SKIP UMS

Chosen (trace scale): JAF + SHJ + AJM **Best:** 765 (AED 765,000/yr gross-benefit proxy) **Units used:** 41 of 45

Greedy: order JAF > UMS > SHJ > AJM picks JAF + UMS + SHJ (AED 720,000/yr) **Gap (DP – greedy):** AED 45,000/yr in gross-benefit proxy

Traceback, greedy comparison, result and service disposition

Start at the bottom-right cell; a cell that differs from the one above means that upgrade was taken.

Step	Item stage	Capacity w	Compare (AED '000)	Decision	Capacity left
1	4 — AJM	45	K[4][45] = 765 vs K[3][45] = 720	TAKE AJM	31
2	3 — SHJ	31	K[3][31] = 585 vs K[2][31] = 495	TAKE SHJ	18
3	2 — JAF	18	K[2][18] = 360 vs K[1][18] = 135	TAKE JAF	4
4	1 — UMS	4	K[1][4] = 0 vs K[0][4] = 0	SKIP UMS	4

Greedy order	Ratio	Units	Fits?	Units left
JAF	2.51	14	yes	31
UMS	1.93	7	yes	24
SHJ	1.72	13	yes	11
AJM	1.31	14	no	11

Result

	Dynamic programming (optimal)	Greedy (best benefit per AED first)
Upgrades funded	JAF (JAFZA) + SHJ (Sharjah) + AJM (Ajman)	JAF (JAFZA) + UMS (Umm Suqeim) + SHJ (Sharjah)
First-year upgrade cost (exact AED)	411,960	344,340
Budget left (exact AED)	38,040	105,660
Modelled annual gross-benefit proxy (AED/yr)	765,000	720,000

Service disposition — replaces the earlier sheet's single combined coverage row

Service disposition (baseline)	Dynamic programming	Greedy
Direct hub-transfer within target (road-network time ≤ 20 min)	2 of 6 — KAR, QUS	2 of 6 — KAR, QUS
Alternative-mode service within target (assumption)	0 of 6	1 of 6 — UMS (motorbike, assumed 18 min from upgrades.csv)
Local-fulfilment upgrade selected (hub travel time unchanged)	3 of 6 — JAF, SHJ, AJM	2 of 6 — JAF, SHJ
Outside selected service plan	1 of 6 — UMS	1 of 6 — AJM
Destinations with a modelled service disposition	5 of 6	5 of 6

Reading. Under DP, 2 of 6 branches (KAR, QUS) meet the proposed 20-minute target by direct hub transfer; JAF, SHJ, AJM receive local-fulfilment upgrades and UMS remains outside the selected DP plan. Five of six destinations have a modelled service disposition, but only two meet the direct-transfer target. Under greedy, UMS is an assumed alternative-mode service (not direct transfer), JAF, SHJ are local fulfilment and AJM is outside the selected plan. A local-fulfilment mini-hub does not make the hub-to-branch journey zero minutes.

DP vs greedy. Greedy takes JAF first (best ratio) and then the cheap UMS motorbike, leaving too little for a further mini-hub; DP skips it so two more mini-hubs fit. DP's gross-benefit proxy is AED 45,000/yr higher — a difference between modelled gross-benefit proxies, not profit; payback, NPV and ROI are not established.

Agreement with the dashboard. Same inputs as the dashboard's fixed baseline run: grid bottom-right K[4][45] = 765; exact-AED DP JAF + SHJ + AJM; brute force over all 16 combinations agrees. Values computed at generation time from the v2.3 code; nothing typed by hand.